

- 7 A particle P is projected with speed $V \text{ m s}^{-1}$ at an angle 75° above the horizontal from a point O on a horizontal plane. It then moves freely under gravity.

(a) Show that the total time of flight, in seconds, is $\frac{2V}{g} \sin 75^\circ$. [2]

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A smooth vertical barrier is now inserted with its lower end on the plane at a distance 15 m from O . The particle is projected as before but now strikes the barrier, rebounds and returns to O . The coefficient of restitution between the barrier and the particle is $\frac{3}{5}$.

(b) Explain why the total time of flight is unchanged. [1]

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- (c) Find an expression for V in terms of g . [7]

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